UNIVERSITY EXAMINATIONS : 2016/2017

EXAMINATION FOR THE DEGREE OF BACHELOR OF BUSINESS INFORMATION TECHNOLOGY

BBIT 304 ADVANCED DATABASES SYSTEMS

MODE: DISTANCE LEARNING

ORDINARY EXAMINATIONS

DECEMBER.2016 **DURATION: 2 HOURS**

INSTRUCTIONS: Answer Question one and any other two questions

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QUESTION ONE [30 MARKS]

a) Briefly explain the following concepts with regard to advanced databases.
   i. B+Tree
   ii. Hash table
   iii. Data warehouse
   iv. Database manager

[8 Marks]

b) A database is defined as a shared collection of logically related files that store organizational data. In this context compare and contrast relational databases and object oriented databases.

[8 Marks]

c) State and explain any **FOUR** advantages of using a prototype in database application development.

[4 Marks]

d) Discuss the consequences of **THREE** of interference problems in concurrently executing transactions in a database.

[6 Marks]

e) Database backup is an important activity during database administration. This exercise involves copying and archiving the data so it may be used to restore the original data after a data loss event. In this background, describe **FOUR** types of database backup.

[4 Marks]

QUESTION TWO [20 MARKS]

a) Study the SQL code below written in MySQL:

```
CREATE TABLE TEACHER
(
TearcherID varchar(100) not null,
Firstname varchar(100),
Lastname varchar(100),
Age Int,
PRIMARY KEY(TeacherID)
);
```

State and explain **TWO** types of integrity used in this code.

[4 Marks]

b) Study the HOTEL table below and answer the questions that follow:

ID	name	location	category
H1	Blue lagoon beach hotel	Blue bay	3 star
H2	Ambre hotel	Palma	3 star
H3	Hilton	Wolmar	5 star
H4	Maritim	Terre Rouge	5 star
H5	Royal palm	Grand Baie	5 Star
H6	Tamassa hotel	Bel Ombre	4 Star

Write SQL statement for each of the following cases:

i. Display all non five star hotels

ii. Update all three-star hotels to 4 star

iii. Display all hotels whose location name ends with letter "e".

iv. Display the hotels in ascending order of the name.

[8 Marks]

c) Describe any **FOUR** activities performed by the database designer during the physical design stage.

[4 Marks]

d) Discuss any **TWO** database access methods you can use in a relational database.

[4 Marks]

QUESTION THREE [20 MARKS]

a) Database tuning is an important activity during database maintenance. Describe **FOUR** guidelines followed in database tuning.

[8 Marks]

b) Over time the databases grow in size and unnecessarily use disk space. Additionally, repeated modifications to the database file may result in data corruption. This risk increases for databases shared by multiple users over a network. Briefly describe any **FOUR** guidelines followed in database repair.

[8 Marks]

c) A database lock is a variable associated with a data item in the database. Describe the difference between binary locks and multi-mode locks.

[4 Marks]

QUESTION FOUR [20 MARKS]

a) Query processing would mean the entire process that involves query translation, optimization, evaluation and finally extraction of data from the database. In this respect discuss the **FIVE** key steps in query processing:

[10 Marks]

b) A database transaction is defined as a sequence of operations that must be executed as a whole taking a consistent database state to another consistent state. Describe **FIVE** states of a database transaction operation(s).

[10 Marks]

QUESTION FIVE [20 MARKS]

a) You have been requested to design a database for a certain football league. The client has provided you with the following requirements:
"The league has several teams. Each team has several players, officials and home grounds. A player can only play for one and only one team. A player has a country of birth but a country can have more than one player in the same or different teams. Team officials have various responsibilities in managing the team. But an official can only attend to a particular team. A team may have one or more home grounds but a particular ground/field belongs to one and only one team".
Using Chen's notation, produce a conceptual database design for this case complete with cardinalities and optionalities

[10 Marks]

b) With respect to logical database design explain the difference between the following pairs of concepts:
   i. Single valued and multi-valued attributes
   ii. Primary key and candidate key
   iii. First normal form and un-normalized form
   iv. Transitive dependency and partial dependency

[4 Marks]

c) Briefly describe the following types of advanced databases.
   i. Temporal databases
   ii. Spatial databases
   iii. Mobile databases

[6 Marks]